

# 1 My solutions for exercises

I'm practicing proof writing as well as problem solving here, so the writing style can vary from one solution to another.

## 1.1 Exercise 2.2.3

### 1.1.1 (a) $a \geq a$ .

Proof. We seek an  $m$  such that  $a = a + m$ . Clearly 0 is such an  $m$ , and hence  $a \geq a$ .

### 1.1.2 (b) If $a \geq b$ and $b \geq c$ , then $a \geq c$ .

Proof. Let  $m$  and  $n$  be natural numbers such that  $a = b + m$  and  $b = c + n$ . Then we have  $a = (c + n) + m$ . By the associative law for addition,  $a = c + (n + m)$ . Therefore  $a \geq c$ .

### 1.1.3 (c) If $a \geq b$ and $b \geq a$ , then $a = b$ .

Proof. Let  $m$  and  $n$  be natural numbers such that  $a = b + m$  and  $b = a + n$ . Then we have  $a = (a + n) + m$  and  $a = a + (n + m)$ , and so  $n + m$  must be zero. Thus  $n = m = 0$  by Corollary 2.2.9. Substituting 0 for  $m$  into  $a = b + m$ , we conclude that  $a = b$ .

1. Lemma. If  $a + m = a$  then  $m = 0$ .

Proof.  $a + m = a = a + 0$ . Using the cancellation law gives  $m = 0$ .

### 1.1.4 (d) $a \geq b$ iff $a + c \geq b + c$ .

Proof.

1. (i) If  $a \geq b$  then  $a + c \geq b + c$ .

Let  $a = b + m$  for some natural number  $m$ . Then  $a + c = (b + c) + m$ , hence  $a + c \geq b + c$ .

2. (ii) If  $a + c \geq b + c$  then  $a \geq b$ .

Since  $a + c \geq b + c$ , there exists a natural number  $m$  such that  $a + c = (b + c) + m$ . Then  $a + c = (b + m) + c$ . By the cancellation law  $a = b + m$ . Hence  $a \geq b$ .

**1.1.5 (e)  $a < b$  iff  $a++ \leq b$ .**

Proof. First, suppose  $a < b$ . Then there exists a natural number  $m$  such that

$$b = a + m$$

Since  $a < b$  implies  $a \neq b$ ,  $m \neq 0$ . Thus by Lemma 2.2.10, there is a natural number  $n$  such that  $n++ = m$ . Hence

$$b = a + (n++) = (a + n)++ = (a++) + n$$

Therefore  $a++ \leq b$ .

Conversely, suppose that  $a++ \leq b$ . Let  $m$  be a natural number such that  $b = (a++) + m$ . Then

$$b = (a++) + m = (a + m)++ = a + (m++)$$

Suppose, for contradiction, that  $a = b$ . Then

$$a = a + (m++)$$

and cancelling  $a$  gives  $m++ = 0$ . But that contradicts Axiom 2.3, which says  $m++ \neq 0$ . Thus  $a \neq b$ .

Since  $b = a + (m++)$ , we conclude that  $a < b$ .

**1.1.6 (f)  $a < b$  iff  $b = a + d$  for some positive number  $d$ .**

Proof. First, assume that  $a < b$ . Then  $b = a + d$  for some natural number  $d$ . Suppose, for contradiction, that  $d = 0$ . then  $b = a + 0 = a$ , contradicting the fact that  $a \neq b$ . Hence  $d \neq 0$ , and we conclude  $d$  is positive.

Conversely, assume that  $b = a + d$  for some positive number  $d$ . If  $a = b$ ,  $d = 0$  by the same argument used in (e), contradicting the fact that  $d \neq 0$ . Thus  $a \neq b$  and we conclude that  $a < b$ .

**1.2 Exercise 2.2.4**

Proposition 2.2.13. Let  $a$  and  $b$  be natural numbers. Then exactly one of the following statements is true:  $a < b$ ,  $a = b$ ,  $a > b$ .

Proof. We first prove that at most one of the statements is true. If  $a = b$ , by definition  $a < b$  and  $a > b$  is false. If  $a < b$  or  $a > b$ , then by definition  $a \neq b$ . Finally, if  $a < b$  and  $a > b$ , then  $a = b$  by Proposition 2.2.12.(c), a contradiction. Hence at most one of the statements is true.

We now prove that at least one of the statements is true. We proceed by induction on  $a$ . Consider the base case  $a = 0$ . Since  $b = 0 + b$ , we have  $0 \leq b$  for all  $b$ . Thus either  $0 < b$  or  $0 = b$ , and the base case is done. Now suppose inductively that  $a < b \vee a = b \vee a > b$ . We need to show that the proposition still holds on  $a++$ . If  $a = b$ , we have  $a++ = b++ = b + 1$ . Since  $a++ \neq b$ ,  $a++ > b$ . If  $a < b$ ,  $a++ \leq b$  by Proposition 2.2.12.(e). Then either  $a++ < b$  or  $a++ = b$ , and in either case we are done. Finally, if  $a > b$  then for some natural number  $m$ ,  $a = b + m$ . Now  $a++ = (b + m)++ = b + (m++)$ . Hence  $a++ > b$ . This closes the induction.

### 1.3 Exercise 2.2.5.

Proposition 2.2.14 (Strong principle of induction). Let  $m_0$  be a natural number, and let  $P(m)$  be a property pertaining to an arbitrary natural number  $m$ . Suppose that for each  $m \geq m_0$ , we have the following implication: If  $P(m')$  is true for all natural numbers  $m_0 \leq m' < m$ , then  $P(m)$  is also true. (In particular, this means that  $P(m_0)$  is true, since in this case the hypothesis is vacuous.) Then  $P(m)$  is true for all natural numbers  $m \geq m_0$ .

The solution became unnecessarily long here because I tried to give a different proof for each lemma.

#### 1.3.1 Lemma a. $a \neq 0 \rightarrow \exists m(N(m) \wedge a = 1 + m)$ .

Proof. Since  $a$  is positive, by Lemma 2.2.10 there exists a natural number  $m$  such that  $a = m++$ . Then we have  $a = m++ = 0 + m++ = (0 + m)++ = (0++) + m = 1 + m$ .

#### 1.3.2 Lemma b. $m \neq 0 \wedge n \neq 0 \rightarrow m + n \neq 1$ .

Proof 1. Suppose  $m + n = 1$ . by Lemma 2.2.10, there exist natural numbers  $p$  and  $q$  such that  $p++ = m$  and  $q++ = n$ . then  $m + n = (p++) + (q++) = ((p + q)++) = 1 = 0++$ . by Axiom 2.4. we infer  $(p + q)++ = 0$ , contradicting axiom 2.3., i.e.,  $(p + q) \neq 0$ . hence  $m + n \neq 1$ .

Proof 2. Suppose  $m + n = 1$ . by Lemma 2.2.10, there exists a natural number  $p$  such that  $p++ = m$ . then  $m + n = (p++) + n = (p + n)++ = 1 = 0++$ . by axiom 2.4 we infer  $p + n = 0$ . but since  $n$  is positive,  $p + n \neq 0$ . By Proposition 2.2.8, a contradiction. Hence  $m + n \neq 1$ .

**1.3.3 Lemma c.**  $a < b++ \wedge a \neq b \rightarrow a < b$ .

Proof 1 (direct). By Proposition 2.2.12.(f),  $b++ = a + n$  for some positive number  $n$ . Since  $n$  is positive, by Lemma a there exists an  $m$  such that  $n = m + 1$ . Now  $b++ = b + 1 = a + n = a + (m + 1) = (a + m) + 1$ . Cancelling 1 from both sides yields  $b = a + m$ . Thus  $b \geq a$ . Since we also have  $a \neq b$ , we conclude that  $b > a$ .

Proof 2 (indirect). Suppose  $a < b$  is false. Then by Proposition 2.2.13,  $a > b$  must hold. Now choose two positive numbers  $n$  and  $m$  such that  $b++ = a + n$  and  $a = b + m$ . Now  $b++ = b + 1 = a + n = (b + m) + n = b + (m + n)$ . Cancelling  $b$  from both sides gives  $m + n = 1$ . But this contradicts Lemma b, since  $m$  and  $n$  are both positive. Hence  $b > a$ .

**1.3.4 Proof**

Proof. Let  $Q(n)$  be the property that  $P(m)$  is true for all  $m_0 \leq m < n$ . We prove by induction that  $Q(n)$  is true for every natural number  $n$ .

If  $n = 0$  then  $Q(0)$  is vacuously true, since there's no  $m$  such that  $m < 0$ .

Now suppose inductively that  $Q(n)$ . Let  $m$  satisfy  $m_0 \leq m < n++$ . We must show  $P(m)$ . We can think of two cases here:  $m = n$  and  $m \neq n$ .

First, consider the case  $m = n$ . Notice that by the inductive hypothesis,  $P(m')$  for all  $m_0 \leq m' < n$ . It follows that  $P(n)$ , i.e.,  $P(m)$ , from the hypothesis of the theorem, and we are done.

Now if  $m \neq n$ , by Lemma c  $m < n$ . Thus by the inductive hypothesis we infer that  $P(m)$ .

This closes the induction, and we have proven that  $Q(n)$  holds for every natural number  $n$ .

Using this result, we now show the conclusion of the theorem, that is,  $P(m)$  for all  $m \geq m_0$ . For every natural number  $m \geq m_0$ ,  $Q(m++)$  holds, since  $m++$  is a natural number. Together with the fact that  $m_0 \leq m < m++$ , we conclude that  $P(m)$ . This demonstrates the strong principle of induction.

**1.4 Exercise 2.2.6.**

Let  $n$  be a natural number, and let  $P(m)$  be a property pertaining to the natural numbers such that whenever  $P(m++)$  is true, then  $P(m)$  is true. Suppose that  $P(n)$  is also true. Prove that  $P(m)$  is true for all natural numbers  $m \leq n$ ; this is known as the principle of backwards induction.

### 1.4.1 Initial solution

1. Lemma. If  $m \leq 0$  then  $m = 0$ .

Proof. Let  $m \leq 0$ . By definition  $0 = m + n$  for some natural number  $n$ . But by Corollary 2.2.9,  $m$  and  $n$  must both be zero. Hence  $m = 0$ .

2. Solution

We induct on  $n$ . We first prove the principle holds on  $n = 0$ . Let  $m$  be a natural number such that  $m \leq 0$ . Then  $m$  must be 0, and from  $P(0)$  we immediately obtain that  $P(m)$ .

Now suppose inductively that the principle holds on  $n$ . We have to prove the principle on  $n++$ . We first obtain the conclusion of the inductive hypothesis. We have  $P(n++)$ , so we infer  $P(n)$  from the hypothesis. we have the requirements of backwards induction on  $n$ , and thus  $P(m)$  is true for every natural number  $m \leq n$ .

Now let  $m$  be a natural number such that  $m \leq n++$ . If  $m = n++$ , then  $P(n++)$ , i.e.,  $P(m)$  and we are done. Otherwise,  $m \leq n$  and from the earlier result we obtain  $P(m)$ . This closes the induction, and we have demonstrated the principle of backwards induction.

### 1.4.2 Solution with revised writing & structure

Using the same lemma as above.

Let  $Q(n)$  be the property that: "if  $P(n)$  is true, then  $P(m)$  is true for all  $m \leq n$ ". We prove  $Q(n)$  for every natural number  $n$ , using induction.

Consider the base case  $n = 0$ . Let  $m \leq 0$ . Then  $m$  must be 0 by the above lemma, and since  $P(0)$ , we conclude that  $P(m)$ .

Suppose inductively that  $Q(n)$ . We have to prove  $Q(n++)$ . From  $P(n++)$  and the definition of  $P$ , we have  $P(n)$ . Thus by the inductive hypothesis,  $P(m)$  for all  $m \leq n$ .

Now let  $m$  be a natural number such that  $m \leq n++$ . If  $m = n++$ , then we substitute  $m$  for  $n++$  in  $P(n++)$ , and we are done. Otherwise,  $m \leq n$  and thus  $P(m)$ . This closes the induction.

## 1.5 The product of two natural numbers is a natural number.

Proof. Let  $n, m$  be natural numbers. We induct on  $n$ , keeping  $m$  fixed.

The base case is obvious, since  $0 \cdot m = 0$ , and 0 is a natural number.

Now suppose inductively that  $n \cdot m$  is a natural number. Then  $(n++) \cdot m = (n \cdot m) + n$  by Definition 2.3.1. Since  $n \cdot m$  and  $m$  are natural numbers,  $(n \cdot m) + n$  is so by the property of addition. This closes the induction.

### 1.6 Exercise 2.3.1.

Lemma 2.3.2 (Multiplication is commutative). Let  $n, m$  be natural numbers. Then  $n \cdot m = m \cdot n$ .

#### 1.6.1 Lemma. $n \cdot 0 = 0$ .

Proof. We prove by induction. The base case  $0 \cdot 0 = 0$  is straightforward from the definition. Now if  $n \cdot 0 = 0$ , then  $(n++) \cdot 0 = (n \cdot 0) + 0 = 0 + 0 = 0$ .

#### 1.6.2 Lemma. $m \cdot (n++) = (m \cdot n) + m$ .

Proof. We use induction on  $m$ . For the base case, by definition of multiplication we have  $0 \cdot (n++) = 0 = 0 + 0 = (0 \cdot n) + 0$ , and we are done.

Now suppose inductively that  $m \cdot (n++) = (m \cdot n) + m$ . Then by definition of multiplication

$$(m++) \cdot (n++) = (m \cdot (n++)) + n++ = (m \cdot n + m) + n++$$

And the properties of addition gives

$$(m \cdot n + m) + n++ = m \cdot n + (m + n)++ = m \cdot n + n + (m++)$$

Finally, applying the definition of multiplication we obtain

$$(m++) \cdot n + m++.$$

Thus  $(m++) \cdot (n++) = (m++) \cdot n + m++$ . This closes the induction.

### 1.6.3 Solution

We induct on  $n$ . Consider the base case  $n = 0$ . Then by definition of multiplication and the first lemma,  $0 \cdot m = 0 = m \cdot 0$ .

Now we prove the inductive step  $(n++) \cdot m = m \cdot (n++)$ . We start from the inductive hypothesis  $n \cdot m = m \cdot n$ . We add  $m$  to both sides and obtain  $(n \cdot m) + m = (m \cdot n) + m$ . By definition, LHS is equal to  $(n++) \cdot m$ . Also, the above lemma says that  $(m \cdot n) + m$  is equal to  $m \cdot (n++)$ . Thus we conclude that  $(n++) \cdot m = m \cdot (n++)$ , and this closes the induction.

### 1.7 Exercise 2.3.2

Lemma 2.3.3 (Positive natural numbers have no zero divisors). Let  $n, m$  be natural numbers. Then  $n \cdot m = 0$  if and only if at least one of  $n, m$  is equal to zero. In particular, if  $n$  and  $m$  are both positive, then  $nm$  is also positive.

Proof. First, suppose that  $n = 0 \vee m = 0$ . If  $n = 0$ , then by Definition 2.3.1  $n \cdot m = 0 \cdot m = 0$ . Otherwise,  $m = 0$ , and  $m \cdot n = 0$  by the same argument. Since multiplication is commutative, that leads to  $n \cdot m = 0$ .

Conversely, suppose that  $nm = 0$ , and for contradiction both  $n$  and  $m$  are positive. Then by Lemma 2.2.10,  $n = p++$  for some natural  $p$ . Now  $n \cdot m = (p++) \cdot m$ , and by definition of multiplication  $n \cdot m = p \cdot m + m$ . Since  $n \cdot m = 0$ , we have  $p \cdot m + m = 0$ . By Corollary 2.2.9 it follows that  $m = 0$ , a contradiction. Thus at least one of  $n$  and  $m$  must be zero.

### 1.8 Proposition 2.3.4

Proposition 2.3.4 (Distributive Law). For any natural numbers  $a, b, c$ , we have  $a(b + c) = ab + ac$  and  $(b + c)a = ba + ca$ .

Proof. Since multiplication is commutative, we only have to show the first identity. We induct on  $a$ , keeping  $b$  and  $c$  fixed.

For the base case  $a = 0$ , we have the following identities:

$$0(b + c) = 0 = 0 + 0 = 0b + 0c$$

Hence the base case is done.

Now suppose inductively that  $a(b + c) = ab + ac$ . Then

$$\begin{aligned} (a++) (b + c) &= a(b + c) + b + c && \text{Definition of multiplication} \\ &= (ab + ac) + b + c && \text{Inductive hypothesis} \\ &= (ab + b) + (ac + c) && \text{Commutativity and associativity of addition} \\ &= (a++)b + (a++)c && \text{Definition of multiplication} \end{aligned}$$

This proves the inductive step, and thus  $a(b + c) = ab + ac$  for all natural numbers  $a, b, c$ .

### 1.9 Exercise 2.3.3

Proposition 2.3.5 (Multiplication is associative). For any natural numbers  $a, b, c$ , we have  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ .

Proof. We use induction on  $a$ . The base case  $a = 0$  is straightforward: both  $(0 \cdot b) \cdot c$  and  $0 \cdot (b \cdot c)$  are equal to 0 by the definition of multiplication.

Now suppose inductively that  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ . We need to show that  $((a++) \cdot b) \cdot c = (a++) \cdot (b \cdot c)$ . The left-hand side is  $(a \cdot b + b) \cdot c$  by definition, which is by distributive law equal to  $(a \cdot b) \cdot c + b \cdot c$ . The right-hand side is  $a \cdot (b \cdot c) + b \cdot c$ . But by the inductive hypothesis  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ , so we conclude that both sides are equal. This closes the induction.

### 1.10 Proposition 2.3.6

Proposition 2.3.6 (Multiplication preserves order). If  $a, b$  are natural numbers such that  $a < b$ , and  $c$  is positive, then  $ac < bc$ .

Proof. By Proposition 2.2.12.(f), there exists a positive number  $m$  such that  $b = a + m$ . Then  $bc = (a + m)c = ac + mc$ . Since  $m$  and  $c$  are both positive,  $mc > 0$  by Lemma 2.3.3. Thus, using Proposition 2.2.12.(f) again, we can conclude that  $bc > ac$ .

### 1.11 Corollary 2.3.7

Corollary 2.3.7 (Cancellation law). Let  $a, b, c$  be natural numbers such that  $ac = bc$  and  $c$  is non-zero. Then  $a = b$ .

Proof. We proceed by contradiction. Suppose  $a \neq b$ . Then by the trichotomy of order, either  $a > b$  or  $a < b$ . If  $a > b$ , by Proposition 2.3.6  $ac > bc$ , which contradicts  $ac = bc$ . We can also obtain the same result when  $a < b$ . Hence  $a = b$ .